TECHNICAL BULLETIN

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EPOXY CURING AGENT 100H
(ECA 100H)

Typical Properties

Characteristic	Typical	Test Method
Molecular Weight (average)	162 - 166	
MHHPA, %	35 Min	D-160
HHPA, %	46.0-54.0	D-160
Specific gravity @ 25° C, g/cc	1.17 - 1.19	D-102 (ASTM D 4052)
Color, Gardner	3 Max	D-150G
Appearance	Clear to yellow liquid	D-104 (Visual)
Viscosity (Brookfield, cps, at 25°C)	50-100	D-150F
Maleic Anhydride, %	0.1 Max	D-160
Acid, %	2.0 Max	D-160B3
Volatiles, ppm	100 Max	D-160A

Applications

ECA 100H is a light colored, low viscosity liquid anhydride epoxy curing agent designed for use in converting epoxy resins to highly crosslinked polymers with excellent physical and electrical properties.

- A replacement for HHPA in most electrical and electronic applications.
- Improves outdoor durability (UV resistant)
- Good alkaline resistance in pipe applications.
- Lower color of cured polymer.

PRODUCT TECHNICAL BULLETIN
ECA 100H

ECA 100H imparts ease of handling due to low viscosity, low toxicity and low volatility. It meets the need for liquid stability under normal plant operating conditions. If the product crystallized during low temperature transit and/or storage, it will return to a liquid at 25°C or less.

When properly formulated with promoters and blended with bisphenol-A, cycloaliphatic, glycidyl ethers or novolac epoxide resins, heat cured systems yield excellent properties in filament wound, pultrusion and laminated forms, as well as, electrical potting, casting, insulating and encapsulating applications.

For more information on the use of anhydrides like ECA 100H as epoxy curing agents, please consult the Technical Bulletin "Formulating Anhydride-Cured Epoxy Systems" available from Dixie Chemical Company.

Formulation Example

ECA 100H was formulated with a variety of epoxy resins. The samples were formulated with 100 parts of epoxy resin, 84-117 parts ECA 100H, and 1 part 1-Methylimidazole. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

PRODUCT TECHNICAL BULLETIN
ECA 100H

Epoxy type	ECA 100H, phr	Viscosity @ 25 °C, cP	Gel Time [1], min	Tg [2], °C
Standard BPA Liquid Epoxy	85	820	33	152
Low Viscosity BPA Liquid	88	610	35	148
Cycloaliphatic Epoxy	117	130	40	222
Epoxy Phenol Novolac	93	730	30	159
Epoxy BPA Novolac	84	2990	34	150
BPF Liquid Epoxy	95	400	35	140

1. Shyodu Hot Pot (100g @ 100°C)
2. Cured 1 hr @ 120°C and 1 hr @ 220°C. Analyzed in DSC: 40°C/min to 250°C

Cure Catalysts

Dixie Chemical Company makes a liquid blend of 60% benzyltriethylammonium chloride and 40% ethylene glycol. Named AP-6G, this liquid can be easily dispersed in liquid epoxy-ECA 100H blends. AP-6G imparts good glass transition temperature (Tg) and extended pot life to anhydride/epoxy formulations, and allows the production of light colored polymers. An example formulation is shown in the table below. Samples were cured for 30 minutes at 100°C, the temperature was raised to 160°C over ten minutes, and the cure was continued for a total of 70 minutes. Gel time and glass transition data for these samples are also shown in the table below.

Parts by Weight ECA 100H	Parts by Weight DGEBA	PHR AP-6G based on ECA 100H	Sunshine Gel Time @ 100 °C (min)	Tg, °C
84	100	2.6	18	146

Handling & Storage

ECA 100H will react with water to form diacids. This is normally undesirable, so ECA 100H should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 100H, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 100H is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.